Projective Reed–Solomon Syndromes Beyond Redundancy Four: Complete First Levels and Coherent Polar Flags

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Abstract

We classify, up to projective-semilinear equivalence, the one-column maximum-distance-separable extensions of projective Reed–Solomon codes at redundancy five over every prime power $q \geq 7$, and give exact projective counts. Equivalently, we classify the deep-hole syndrome directions in the first case left open by the classification through redundancy four. A syndrome determines a Hankel linear system of binary forms and is deep exactly when that system contains no completely split squarefree member. The generic redundancy-five case is controlled by an integral off-diagonal fiber-square curve of arithmetic genus one.

For higher redundancy we introduce coherent polar flags: iterated contractions in which every selected factor remains a marked forbidden root. Separating contained components from transverse point counts gives complete all-field syndrome classifications at redundancies six and seven; the redundancy-seven split-free list is a deep-hole classification whenever the covering radius is six, in particular for $q \geq 11$. As a first fixed-level application, the three-marker argument leaves only the persistent tangent and conjugate-secant families at redundancy eight for $q \geq 43$.

Index Terms

Algebraic coding theory, deep holes, finite fields, maximum-distance-separable extensions, normal rational curves, projective Reed–Solomon codes.

I. Introduction

This paper has two principal contributions. First, it gives the complete all-field classification at redundancy five, the first case left open by the known classification through redundancy four. Second, it introduces coherent polar flags: a marked contraction method that propagates splitting information to higher redundancies without forgetting the factor needed for a squarefree lift. This gives complete all-field syndrome classifications at redundancies six and seven and, as a fixed-level application, the high-field classification at redundancy eight. We use deep-hole syndromes because their Hankel systems make both the base classification and the propagation mechanism visible.

Let $\mathrm{PRS}(k)$ denote the projective Reed–Solomon code of length $q+1$ obtained by evaluating homogeneous binary forms of degree $k-1$ on $\mathbb{P}^1(\mathbb{F}_q)$. If the redundancy is

$$r = q + 1 - k,$$

a parity-check matrix has as columns the $\mathbb{F}_q$-points of the degree-$(r-1)$ normal rational curve

$$\mathcal{R}_{r-1} = \{(1, t, \ldots, t^{r-1}) : t \in \mathbb{F}_q\} \cup \{(0, \ldots, 0, 1)\} \subset \mathbb{P}^{r-1}.$$

Call a projective syndrome direction split-free if it is not contained in the span of $r-2$ distinct columns. When $\rho(\mathrm{PRS}(q+1-r)) = r-1$, the split-free directions are precisely the projective syndrome directions of deep-hole cosets. The logical separation used throughout the paper is

$$\boxed{\text{split-free classification} \ + \ \rho(\mathrm{PRS}(q+1-r)) = r-1 \quad \Longrightarrow \quad \text{deep-hole classification.}}$$

The first term is proved geometrically; the covering-radius equality is a separate coding-theoretic input and is never inferred from a finite split-free census.

The correspondence between deep holes, one-symbol MDS extensions, and uncovered points of a normal rational curve was developed in [1], [2]. Zhang, Wan, and Kaipa classified projective Reed–Solomon deep-hole directions through redundancy four and identified redundancy five as the next case [2]. Our first theorem gives the corresponding all-field redundancy-five classification.

Theorem I.1 (Complete redundancy-five classification). Let $q \geq 7$ be a prime power. The covering radius of $\mathrm{PRS}(q-4)$ is 4. Its projective deep-hole syndromes are exactly the following disjoint families:

(i) the tangent family T, of size $q(q+1)$;
(ii) the conjugate-secant family S, of size $q(q^2-1)/2$;

(iii) if $\operatorname{char}\mathbb{F}_q \neq 3$, the rational osculating-pair family O^+ when $q \equiv 2 \pmod 3$, of size $q(q+1)/2$, or the conjugate osculating-pair family O^- when $q \equiv 1 \pmod 3$, of size $q(q-1)/2$;
(iv) in characteristic three, one $\mathrm{PGL}_2(q)$-fixed nucleus point and one wild Artin–Schreier orbit W of size $(q^2-1)/2$;
(v) the sporadic $\mathrm{PGL}_2(q)$-orbits in Table IV, occurring exactly for $q \in \{7, 8, 9, 11, 13, 17, 19\}$.

There are no other deep-hole syndromes. The three corresponding nonsporadic totals are

$$\frac{q^2(q+3)}{2}, \qquad \frac{q(q+1)(q+2)}{2}, \qquad \frac{q^3+3q^2+q+1}{2},$$

according as $q \equiv 1 \pmod 3$ with characteristic not three, $q \equiv 2 \pmod 3$, or $\operatorname{char}\mathbb{F}_q = 3$.

The main mechanism is a characteristic-free Hankel dictionary. A syndrome at redundancy r annihilates a linear system of binary forms of degree $r-2$. It is split-free if and only if this system contains no completely split squarefree form; it is a deep-hole direction when, in addition, $\rho(\mathrm{PRS}(q+1-r)) = r-1$. At redundancy five the system is a pencil of cubics. Its induced degree-three map of projective lines is either cyclic or has geometric monodromy S_3. Cyclic descent gives the arithmetic families in Theorem I.1; in the S_3 case the off-diagonal fiber square is an integral curve of arithmetic genus one, and the singular-curve Weil bound [3] forces a split fiber for $q \geq 23$. The certified finite computations in Section VIII handle the remaining fields.

Starting at redundancy six, the Hankel system is no longer a pencil. The useful replacement is the coherently parameterized line of contractions obtained by choosing one prospective root. Iteration produces a polar flag. Geometrically, naïve contraction keeps the lower divisor but forgets which point of $\mathbb{P}^1$ was removed. If the lower divisor passes through that point, putting the point back creates a double point rather than a squarefree divisor. The polar flag retains the removed point as a forbidden marker, exactly the datum needed for lifting. This pointed coherence is essential: at $q = 19$ an individual contracted fiber has no split witness avoiding its marker, but no consecutive-row polar line is contained in its orbit.

The resulting structural theorem has three outputs.

Theorem I.2 (Classification results and explicit gates through eight). The following statements hold.

(a) For every prime power $q \geq 7$, the projective deep-hole directions of $\mathrm{PRS}(q-5)$ are exactly the persistent tangent and conjugate-secant families, the exceptional orbits listed for $q = 7, 8, 9, 11, 13$, and, when $q = 2^m$ with odd $m \geq 5$, one characteristic-two nucleus orbit.
(b) The displayed finite domains through $q = 32$ have the certified split-free classification stated in Section VI. Together with the structural theorem, these finite classifications give the all-field result: for every $q \geq 13$ the split-free locus consists of the persistent families and, when $q = 2^m$ with m odd, one central nucleus point. This is a deep-hole classification only when $\rho(\mathrm{PRS}(q-6)) = 6$, in particular for $q \geq 11$.
(c) For $q \geq 43$, the projective deep-hole directions of $\mathrm{PRS}(q-7)$ are exactly the persistent tangent and conjugate-secant families, of total size $q(q+1)^2/2$.

At redundancy r, the semilinear orbits in the persistent conjugate-secant family are parameterized by

$$T/T^{r-1} \quad \text{modulo inversion and Frobenius,}$$

where $T = \{z \in \mathbb{F}_{q^2}^\times : z^{q+1} = 1\}$ is the norm-one torus and Frobenius is coefficient Frobenius. On the tangent family the unipotent coordinate transforms as $z \mapsto z + (r-1)u$; consequently its orbit decomposition changes precisely when $\operatorname{char}\mathbb{F}_q$ divides $r-1$.

The three parts have different scopes. Parts (a) and (b) are unconditional all-field syndrome classifications, but part (b) becomes a deep-hole classification only when $\rho(\mathrm{PRS}(q-6)) = 6$. Part (c) is unconditional in its displayed high-field range and makes no classification claim below its threshold. Table II places the split-free result, radius gate, deep-hole conclusion, and remaining field gap in separate columns for each redundancy.

a) Relation to previous work.: The tangent and conjugate-secant families and the coding/normal-rational curve dictionary come from [1], [2]; the covering-radius input in the high-rate range is due to Seroussi and Roth [4]. The broader arc/MDS and code-automorphism background is surveyed or established in [5], [6]. Earlier explicit-family results for projective and generalized projective Reed–Solomon codes include [7], [8], [9]; the recent extension framework of [10] further clarifies the deep-hole/MDS-extension interface. These works construct families, treat punctured or lower-redundancy regimes, or analyze the extension criterion; they do not supply the complete redundancy-five cubic-pencil exhaustion proved here. Classical apolarity, catalecticants, and binary-form rank strata are treated in [11], [12]; our use of divided powers keeps the small-characteristic contraction formulas explicit. Factorization statistics in linear families [13], normal-rational-curve nuclei [14], and the general Frobenius/monodromy semantics of splitting families [15] are inputs. The coding-theory references above identify deep syndromes with uncovered points of a normal rational curve and settle the lower-redundancy families; they do not classify the redundancy-five cubic pencils. The orbit-theoretic and factorization references supply normal forms and distribution tools, but not the coherent marked contraction used here. The nucleus results locate characteristic-dependent kernels, while the present work computes which of their coherent lifts meet the PRS splitting incidence. Accordingly, this paper's contributions are separated into the redundancy-five through seven classifications, the conditional transverse induction mechanism, and the first fixed-level high-field application. Claim-by-claim novelty and dependency boundaries are recorded in the electronic ledger.

b) Organization.: Section II first gives a proof-free guide to the argument, and Section III gives the Hankel dictionary. Section IV proves Theorem I.1. Sections V–VII develop coherent polar induction and prove Theorem I.2. Section VIII states the computational and formal trust boundary, and Section IX records the exact open boundary. Companion papers will treat the redundancy-nine slice analysis and the logically separate ordered-Hessian and Lucas-carrier geometry.

II. Guide to the proof

This section is a proof-free guided tour. Its purpose is to install the information flow used throughout the paper before coordinates, finite exceptional tables, or formal trust boundaries intervene.

a) Syndrome to splitting system.: A syndrome direction $[f] \in \mathbb{P}(\Gamma^{r-1}E)$ determines its Hankel annihilator $W_f \subset \operatorname{Sym}^{r-2} E^\vee$. A product of $r-2$ distinct rational linear factors lies in W_f exactly when f lies in the span of the corresponding normal-rational-curve columns. Thus the geometric classification begins with split-free systems. The separate covering-radius theorem is what promotes split-free directions to code deep holes; the dictionary alone does not.

b) The redundancy-five base cover.: At redundancy five, W_f is a pencil of binary cubics. Its root incidence is a degree-three map of projective lines. The off-diagonal fiber square separates the cyclic cases, controlled by Frobenius on a deck transformation, from the S_3 case, controlled by rational points on a geometrically integral $(2,2)$ curve. This is the bottom cover used at every later level.

c) Why independent lower fibers fail.: At higher redundancy, choosing one lower witness forgets which factor was removed. If that lower witness contains the chosen factor, then multiplication restores it with multiplicity two rather than producing a squarefree form. The lost datum is therefore not auxiliary: it is the exact obstruction to lifting.

d) Coherent replacement and the fork.: Choose a prospective root λ, contract f to $\iota_\lambda f$, and retain λ as a marked forbidden root. Iteration gives an ordered polar flag. Its first-polar line has two qualitatively different behaviors. If it is transverse to the lower bad locus and collision schemes, only finitely many parameters must be removed and a point count produces an escape. If it is scheme-theoretically contained, the point-count argument says nothing: ordinary catalecticant components and components created in small characteristic require a separate level-specific classification. Section V introduces the terms persistent and modular carriers only when that distinction enters the proof.

e) Lifting.: On the transverse branch, a rational point of the lower ordered splitting cover outside the branch, diagonal, old-marker, new-marker, and self-collision divisors produces a split squarefree lower witness avoiding all markers. Multiplying by the marked factors then lifts it squarefreely to W_f. Figure 1 records this sequence and the contained/transverse fork.

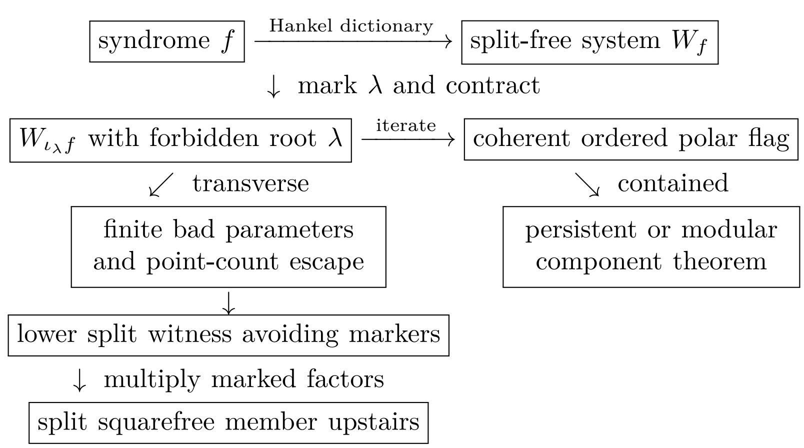

Fig. 1. Information flow of coherent polar induction. Solid arrows are the statements proved in the degrees treated here. The contained branch is not a consequence of the transverse sieve.

TABLE I
Mathematical dependency map.

Result	Printed geometric argument	Imported input
R5	Hankel dictionary; gcd strata; cyclic descent; integral $(2, 2)$ fiber square and point bound	lower tangent/secant families; high-rate radius theorem; singular-curve bound
R6	one-marker polar line; ordinary and binary contained components; transverse escape	R5 lower package and radius gate
R7	two-marker contraction; contained rank-two theorem; transverse escape	R5/R6 lower packages; separate covering-radius result
R8	three-marker cover; explicit contained and collision calculations; point-count budget	proved lower packages

TABLE II
Classification status by redundancy. "Unconditional" refers to the displayed field range. A split-free classification becomes a deep-hole classification only after the separate radius gate.

Red.	Field range	Split-free geometry	Deep holes/MDS extensions	Remaining gap
5	$q \geq 7$	Complete, unconditional	Complete, unconditional; $\rho = 4$	None
6	$q \geq 7$	Complete, unconditional	Complete, unconditional; $\rho = 5$	None
7	$q \geq 7$	Complete, unconditional	Complete for $q \geq 11$; conditional on $\rho = 6$ at $q = 7, 8, 9$	Radius at $q = 7, 8, 9$
8	$q \geq 43$	Exactly the tangent and conjugate-secant families, unconditional in range	Complete, unconditional in range	Fields $7 \leq q < 43$

f) Referee roadmap.: The shortest route to the headline R5 theorem is Lemma III.1 and Corollary III.2, followed by the radius gate and gcd strata in Propositions IV.1–IV.3, the cubic incidence in Proposition IV.4, the cyclic and S_3 analyses in Lemmas IV.5–IV.6, and the finite bridge in Proposition IV.7. Tables IV and V make the bounded-field output locally identifiable and numerically auditable.

The later R6 and R7 proofs use the R5 splitting cover as their bottom package; R8 then uses those proved lower packages recursively. Section V may be read as the abstract mechanism linking these levels. Running the supplement is required only to reproduce the finite bridges, canonical orbit records, and regression or formal checks identified in Section VIII; it is not a substitute for any printed geometric argument.

The proofs now follow these arrows in order: dictionary, redundancy-five base cover, polar construction, contained components, transverse point count, and fixed-level applications. This is a scope boundary, not a second theorem inventory.

III. The syndrome–Hankel dictionary

Let E be a two-dimensional vector space. We use divided powers for syndromes and ordinary symmetric powers for root forms. This avoids invalid binomial rescalings in small characteristic. For $f = (a_0 : \cdots : a_{r-1}) \in \mathbb{P}(\Gamma^{r-1}E)$, write $e_0, \ldots, e_{r-1}$ for the divided-power coordinate basis and $\nu(t) = (1, t, \ldots, t^{r-1})$, with $\nu(\infty) = e_{r-1}$. The perfect divided-power/symmetric-power pairing is characterized by coefficient extraction. Define

$$H_f = \begin{pmatrix} a_0 & a_1 & \cdots & a_{r-2} \\ a_1 & a_2 & \cdots & a_{r-1} \end{pmatrix}, \qquad W_f = \ker H_f \subset \operatorname{Sym}^{r-2} E^\vee.$$

Lemma III.1 (Hankel criterion). Let $\lambda_1, \ldots, \lambda_{r-2}$ be distinct points of $\mathbb{P}(E^\vee)(\mathbb{F}_q)$ and put $g = \prod_i \lambda_i$. Then

$$f \in \operatorname{span}\big(\nu(\lambda_1), \ldots, \nu(\lambda_{r-2})\big) \iff g \in W_f.$$

The same statement with repeated factors and Hasse derivatives gives the corresponding osculating spans. Both statements commute with $\mathrm{P\Gamma L}_2(q)$.

Proof. On an affine chart, g is the characteristic polynomial of an order-$(r-2)$ recurrence. The length-r solutions are spanned by the geometric sequences $\nu(\lambda_i)$; their independence is the Vandermonde determinant. A root at infinity gives the reversed-chart recurrence. For repeated roots, the confluent Vandermonde basis is formed by Hasse derivatives, so the same argument works in every characteristic. Equivariance follows because substitution transports both the normal rational curve and the recurrence. □

Corollary III.2 (Split-free criterion). Assume $\rho(\mathrm{PRS}(q + 1 - r)) = r - 1$. A projective syndrome f is deep if and only if W_f contains no completely split squarefree form of degree $r - 2$ over $\mathbb{F}_q$.

TABLE III
Notation used from the Hankel dictionary onward.

Symbol	Meaning
q	field size; all numerical thresholds are thresholds on prime powers
k	code dimension in $\mathrm{PRS}(k)$
$r = q + 1 - k$	redundancy
$n = r - 1$	divided-power degree of a syndrome representative
E	a two-dimensional vector space over the current base field
$f \in \Gamma^n E$	divided-power representative of a projective syndrome
$W_f \subset \mathrm{Sym}^{n-1} E^\vee$	Hankel/apolar witness system annihilated by f
g, δ, κ	genus, deletion degree, and point-bound correction
b, c	transverse bad-locus and marker-collision degrees

We call the latter property split-free even when the covering-radius hypothesis has not been established. Thus every deep syndrome in the radius-$r-1$ range is split-free, while a small-field split-free census is not silently promoted to a code deep-hole classification. We call a direction shallow when W_f does contain a completely split squarefree form.

The first locus that persists under contraction is already visible here. If every member of W_f has a common quadratic factor, then that factor is either a rational square, an irreducible quadratic, or a product of two distinct rational factors. These are respectively the tangent deep family, the conjugate-secant deep family, and a shallow rational secant. The common factor cannot grow with redundancy: since $\dim W_f = r - 3$,

$$r - 3 \le r - 1 - \deg \gcd(W_f),$$

so its degree is at most two.

A. Coding consequences and recognition

The dictionary has two immediate coding consequences. First, a split-free direction is an uncovered point of the normal rational curve and hence gives a one-column MDS extension whenever the stated covering-radius gate applies. The classification theorems therefore give exact projective deep-hole counts, not only normal forms. Second, they give recognition procedures. From a syndrome one forms H_f, computes W_f, and tests the listed geometric families: a common quadratic factor detects the persistent tangent and conjugate-secant families; the relevant contraction kernel detects a modular nucleus; and the finite exceptional fields are decided by the canonical orbit representatives printed in Table IV and recorded in the supplement.

For each fixed redundancy treated here, constructing and row-reducing the Hankel matrix and performing the listed family-membership tests costs $O(r^3)$ field operations. This count treats factorization of the displayed bounded-degree forms and access to the exceptional orbit tables as fixed-degree work. It excludes field construction, bit complexity, and the one-time generation and certification of those tables; an implementation may also charge separately for its chosen orbit-normalization routine. Thus the bound describes the online linear-algebra stage at the fixed redundancies proved here. It is not an end-to-end uniform arbitrary-r complexity claim, a general decoder, or a preprocessing bound. A direct witness search would instead inspect the projective members of the $(r-3)$-dimensional system W_f and factor degree-$(r-2)$ forms; coherent polar induction is the mechanism that replaces that ambient search in the levels proved below.

IV. Redundancy five: exceptional cubic covers

Here $f = (a_0 : \cdots : a_4)$ and W_f is the pencil annihilated by

$$\begin{pmatrix} a_0 & a_1 & a_2 & a_3 \\ a_1 & a_2 & a_3 & a_4 \end{pmatrix}.$$

These are divided-power coordinates; the coefficients of a cubic in W_f are ordinary symmetric-power coordinates. Factoring the formal quartic $\sum a_i T^i U^{4-i}$ would therefore be noninvariant in characteristics two and three.

Proposition IV.1 (Radius gate). For every prime power $q \ge 7$,

$$\rho(\mathrm{PRS}(q-4)) = 4.$$

Proof. The coding/arc criterion identifies radius four with completeness of the quartic normal rational curve. Seroussi–Roth prove completeness when

$$k \ge \left\lfloor \frac{q-1}{2} \right\rfloor$$

apart from the even-field dimensions $k = 2, q-2$ [4]. Here $k = q-4$, the inequality holds for $q \geq 7$, and neither exceptional dimension occurs. Thus Corollary III.2 applies. The finite certificate independently checks the same conclusion only in its declared finite domain; it is not the all-field radius proof. □

A. Gcd strata

Proposition IV.2 (Quadratic gcd). If $\deg \gcd(W_f) = 2$, then

$$W_f = Q\langle T, U\rangle.$$

If Q is split squarefree, f is shallow. If Q is irreducible, f belongs to the conjugate-secant family; if $Q = \lambda^2$, it belongs to the tangent family. Conversely every point in those families has the displayed gcd. Their deep-family sizes are

$$|\mathrm{T}| = q(q+1), \qquad |\mathrm{S}| = \frac{q(q^2-1)}{2}.$$

Proof. The equality follows by dimension. A split Q and a linear cofactor avoiding its two roots give a split squarefree cubic. An irreducible Q survives in every member, while a square Q forces a repeated root in every member. Lemma III.1, including its confluent and conjugate-root forms, identifies the corresponding syndromes with the secant and tangent spans. The sizes are the established tangent/conjugate-secant counts from the lower-redundancy classification [1], [2]. □

Proposition IV.3 (Linear gcd). Suppose $\deg \gcd(W_f) = 1$. Then f is shallow for $q \geq 8$. At $q = 7$ the same conclusion is part of Certificate R5.

Proof. Write $W_f = \lambda_0 V$ with V a basepoint-free quadratic pencil. If its degree-two map $\psi : \mathbb{P}^1 \to \mathbb{P}^1$ is separable, its rational deck involution ι gives the graph

$$\Gamma_\iota(\mathbb{F}_q) = \{(t, \iota(t)) : t \in \mathbb{P}^1(\mathbb{F}_q)\},$$

which has exactly $q+1$ rational points. A safe union bound deletes at most two fixed graph points, at most four graph points over the branch fibers, and at most the two graph points $(\lambda_0, \iota(\lambda_0))$ and $(\iota(\lambda_0), \lambda_0)$. Thus fewer than $q+1$ graph points are deleted when $q \geq 8$. A surviving point gives a split quadratic with two distinct roots avoiding λ_0, and hence a split squarefree cubic in W_f. In characteristic two an inseparable quadratic pencil is equivalent to $\langle T^2, U^2\rangle$; the two overlapping Hankel equations then force f onto the normal rational curve, contrary to the rank-two pencil hypothesis. □

B. The trivial-gcd cubic cover

Choose a basis c_1, c_2 of a trivial-gcd pencil and put

$$X_f = \{([u:v], t) : uc_1(t) + vc_2(t) = 0\} \subset \mathbb{P}^1 \times \mathbb{P}^1.$$

Proposition IV.4 (Cubic incidence and off-diagonal fiber square). The curve X_f is geometrically integral of bidegree $(1,3)$ and arithmetic genus zero. Projection to the root coordinate is birational, while projection to the pencil coordinate gives a degree-three morphism

$$\phi_f : \mathbb{P}^1 \longrightarrow \mathbb{P}^1.$$

If ϕ_f is separable, division of

$$c_1(t_1)c_2(t_2) - c_2(t_1)c_1(t_2)$$

by the diagonal bracket defines a residual curve $Y_f \subset \mathbb{P}^1 \times \mathbb{P}^1$ of bidegree $(2,2)$ and arithmetic genus one. Its geometric components are the geometric-monodromy orbits on ordered pairs of distinct roots.

Proof. A component of X_f of bidegree $(0,e)$ would be a common factor of c_1, c_2; hence none exists. The bidegree genus formula gives genus zero. A root determines the unique pencil member containing it, so the root projection is birational. For a separable map the diagonal factor in the fiber-square determinant has multiplicity one. Removing its bidegree $(1,1)$ from the fiber square leaves bidegree $(2,2)$, whose arithmetic genus is one. The standard fiber-square/monodromy correspondence identifies its components with orbits on distinct ordered sheets. □

Lemma IV.5 (Cyclic stratum). Suppose ϕ_f is geometrically cyclic.

(i) In characteristic different from three there are two ramification points. If they are rational, the corresponding osculating-plane intersection is deep exactly when $q \equiv 2 \pmod 3$; if they are conjugate, it is deep exactly when $q \equiv 1 \pmod 3$.

(ii) In characteristic three all osculating planes meet in $n = (0:0:1:0:0)$, whose pencil is $\langle T^3, U^3\rangle$; this fixed point is deep.

(iii) The remaining wild characteristic-three pencils have the form $\langle U^3, T^3 + aTU^2\rangle$. They are deep exactly when $-a$ is nonsquare, forming one orbit of size $(q^2-1)/2$.

Proof. Assume first that the characteristic is not three. Riemann–Hurwitz gives two totally ramified points t_1, t_2. The confluent Hankel criterion places f in $\mathrm{Osc}(t_1) \cap \mathrm{Osc}(t_2)$; after transporting the pair to $\{0, \infty\}$, the pencil is $\langle T^3, U^3\rangle$ and a geometric deck generator is $t \mapsto \zeta t$.

For a rational ramification pair, Frobenius fixes the points and the deck map descends exactly when $\zeta^q = \zeta$, namely $q \equiv 1 \pmod 3$. Thus its complement is deep exactly for $q \equiv 2 \pmod 3$. The setwise stabilizer of the ordered normal form is the split dihedral group of order $2(q-1)$, so the orbit size is

$$\frac{q(q^2-1)}{2(q-1)} = \frac{q(q+1)}{2}.$$

For a conjugate pair Frobenius interchanges the fixed points and inverts the multiplier; descent is instead $q \equiv 2 \pmod 3$. The nonsplit dihedral stabilizer has order $2(q+1)$ and the orbit size is $q(q-1)/2$.

Now suppose the characteristic is three. The second Hasse derivative of

$$\nu(t) = (1, t, t^2, t^3, t^4)$$

is $(0, 0, 1, 3t, 6t^2) = e_2$, including in the reversed infinity chart. Thus every osculating plane contains $n = [e_2]$. Substitution shows that n is $\mathrm{PGL}_2(q)$-fixed and $W_n = \langle T^3, U^3\rangle$, so it is deep.

A remaining separable cyclic cover has one wild totally ramified point. Sending it to infinity and comparing coefficients under a nontrivial translation gives

$$W_f = \langle U^3, T^3 + \alpha T^2 U + \beta T U^2\rangle, \qquad \alpha = 0, \quad \kappa^2 = -\beta.$$

It is therefore represented by

$$f_a = e_2 - a e_4, \qquad W_{f_a} = \langle U^3, T^3 + aTU^2\rangle.$$

The affine members are $z^3 + az + s$. Such a member has three rational roots exactly when the additive map $z \mapsto z^3 + az$ has a nonzero rational kernel, equivalently when $-a$ is a square. Hence the nonsquare parameters form the deep orbit. Its stabilizer is $t \mapsto \pm t + \beta$, of order $2q$, and orbit–stabilizer gives $(q^2-1)/2$. ☐

Lemma IV.6 (S_3 stratum). If ϕ_f has geometric monodromy S_3 and $q \geq 23$, then the pencil contains a completely split squarefree cubic.

Proof. By Proposition IV.4, components of Y_f correspond to monodromy orbits on distinct ordered roots. The S_3 action is two-transitive, so Y_f is geometrically integral and has arithmetic genus one. Aubry–Perret gives

$$\#Y_f(\mathbb{F}_q) \geq q - 2\sqrt{q}.$$

The diagonal costs

$$Y_f \cdot \Delta = (2,2) \cdot (1,1) = 4.$$

Riemann–Hurwitz gives different degree four for the separable cubic map, hence at most four branch values. A singular cubic has at most two distinct roots and at most two ordered off-diagonal pairs, so the branch deletion costs at most eight. If $q - 2\sqrt{q} > 12$, a rational off-diagonal unramified point remains. It gives two distinct rational roots of a squarefree cubic, whose third root is rational. The inequality holds for every prime power $q \geq 23$. ☐

Proposition IV.7 (Exhaustion and finite bridge). The preceding cases exhaust every redundancy-five pencil for $q \geq 23$. Certificate R5 closes exactly

$$q \in \{7, 8, 9, 11, 13, 16, 17, 19\}.$$

It finds sporadic S_3-stratum orbits exactly at $q = 7, 8, 9, 11, 13, 17, 19$ and none at $q = 16$.

Proof. The gcd degree is zero, one, or two. Propositions IV.2 and IV.3 settle the positive-gcd cases. In the trivial-gcd case the cubic map is inseparable or separable. The inseparable case occurs only in characteristic three and the overlap equations give the all-cube nucleus pencil. A separable transitive cubic has geometric monodromy C_3 or S_3, settled by Lemmas IV.5 and IV.6. The stated finite residue, canonical representatives, stabilizers, histograms, and Frobenius links are exactly the domain of Certificate R5. ☐

TABLE IV: Complete sporadic orbit inventory at redundancy five. A representative $[a_0, \dots, a_4]$ is in the divided-power syndrome coordinates of Section III. The last column records the nontrivial Frobenius fusion; a dash means the orbit is fixed.

q	representative	orbit size	stabilizer	Frobenius
7	[0,1,0,0,3]	28	12	–
7	[0,1,0,0,2]	112	3	–
7	[0,0,1,0,3]	168	2	–
7	[0,1,0,3,2]	168	2	–
7	[1,0,0,2,1]	168	2	–
8	[0,1,1,0,2]	252	2	$\to [0, 1, 1, 0, 4]$
8	[0,1,1,0,4]	252	2	$\to [0, 1, 1, 0, 6]$
8	[0,1,1,0,6]	252	2	$\to [0, 1, 1, 0, 2]$
9	[1,0,0,1,2]	180	4	–
9	[0,1,0,4,1]	360	2	$\leftrightarrow [0, 1, 0, 4, 4]$
9	[0,1,0,4,4]	360	2	$\leftrightarrow [0, 1, 0, 4, 1]$
11	[1,0,0,1,10]	330	4	–
11	[1,0,0,1,1]	660	2	–
13	[0,1,0,0,4]	182	12	–
13	[1,0,0,4,7]	546	4	–
17	[1,0,0,1,5]	1224	4	–
19	[0,1,0,0,4]	570	12	–

The certificate additionally records the complete cubic-pencil member histogram of each representative. Repeated orbit sizes in Table IV are therefore visibly distinguished without requiring a working-directory file.

TABLE V
Independent R5 completeness totals. The first equality is the direct syndrome enumeration versus the orbit–stabilizer sum; the second decomposes the same total into the formula in Theorem I.1 and the sporadic rows above.

q	direct = orbit sum	nonsporadic	sporadic
7	$889 = 889$	245	644
8	$1116 = 1116$	360	756
9	$1391 = 1391$	491	900
11	$1848 = 1848$	858	990
13	$2080 = 2080$	1352	728
16	$2432 = 2432$	2432	0
17	$4131 = 4131$	2907	1224
19	$4541 = 4541$	3971	570

V. Coherent polar induction

A. Spaces, markers, and bad schemes

Fix a field k, a two-dimensional k-space E, and

$$S_n = \mathbb{P}(\Gamma^n E).$$

For $\lambda \in E^\vee$, define divided-power contraction by

$$\langle \iota_\lambda f, g \rangle = \langle f, \lambda g \rangle.$$

Then

$$g \in W_{\iota_\lambda f} \quad \Longleftrightarrow \quad \lambda g \in W_f. \tag{1}$$

The lower witness lifts squarefreely only when $\lambda \nmid g$. Thus the factor selected for contraction must remain as a marked forbidden root.

Definition V.1 (Carriers). A carrier is a closed locus on which the transverse splitting argument degenerates. A persistent carrier is an ordinary catalecticant component propagated by contraction; a modular carrier is an additional component created by a positive-characteristic contraction kernel.

Definition V.2 (Contraction and marker spaces). Let $C_f : E^\vee \to \Gamma^{n-1}E$ be $\lambda \mapsto \iota_\lambda f$, and put $U_f = \mathbb{P}(E^\vee) \setminus \mathbb{P}(\ker C_f)$. The first-polar rational map is

$$\Phi_f : U_f \longrightarrow S_{n-1}, \qquad [\lambda] \longmapsto [\iota_\lambda f].$$

Its image is a point or a line. For $j \geq 1$, let

$$\mathrm{Conf}_j(\mathbb{P}(E^\vee)) = \{(\lambda_1, \ldots, \lambda_j) : \lambda_i \neq \lambda_\ell \text{ for } i \neq \ell\}.$$

The ordered marker space $U_{f,j}$ is the open subset on which every successive contraction is nonzero. The universal j-step polar flag is the map

$$(\lambda_1, \ldots, \lambda_j) \longmapsto \big([\iota_{\lambda_1} f], \ldots, [\iota_{\lambda_j} \cdots \iota_{\lambda_1} f]\big).$$

In an affine coordinate r the first map is

$$\Phi_f(r) = (a_1 - ra_0, \ldots, a_n - ra_{n-1}), \qquad \Phi_f(\infty) = (a_0, \ldots, a_{n-1}).$$

Example V.3 (The complete R5-to-R6 propagation at $q = 29$). Work over $\mathbb{F}_{29}$ at redundancy six and take

$$f = (6 : 1 : 0 : 0 : 5 : 1).$$

The Hankel kernel is the net of quartics

$$W_f = \ker \begin{pmatrix} 6 & 1 & 0 & 0 & 5 \\ 1 & 0 & 0 & 5 & 1 \end{pmatrix} = \langle (0,0,1,0,0), (24,1,0,1,0), (28,1,0,0,1) \rangle .$$

The displayed basis has no common factor, so the persistent test does not decide the syndrome. Choose the first marker $r = 0$. Then

$$\iota_0 f = (1 : 0 : 0 : 5 : 1).$$

This is now a redundancy-five syndrome, and its full cubic pencil is

$$W_{\iota_0 f} = \langle (0,1,0,0), (25,0,1,24) \rangle .$$

The member with pencil parameter $[9 : 28]$ is

$$h(t) = 4 + 9t - t^2 + 5t^3 \in W_{\iota_0 f},$$

because the two Hankel products are $4 + 5 \cdot 5 = 0$ and $5(-1) + 5 = 0$ in $\mathbb{F}_{29}$. Direct factorization gives

$$h(t) = 5(t-2)(t-6)(t-27).$$

None of these roots is the marked root 0. Equation (1) therefore lifts h to

$$t\, h(t) = (0, 4, 9, 28, 5) \in W_f,$$

a squarefree quartic with roots $0, 2, 6, 27$. The Hankel criterion now places f in the span of the corresponding four normal-rational-curve columns, so f is shallow. This one calculation exhibits every interface in the R5-to-R6 propagation: the upper Hankel net, the ordinary common-factor test, contraction to the complete R5 cubic pencil, selection on its splitting cover, marker avoidance, and squarefree lifting. In the general proof the R5 curve argument supplies such a pencil parameter after deleting the marked fibers; the displayed parameter $[9 : 28]$ is one complete instance of that step.

Definition V.4 (Lower splitting package). At level (n, j) a lower splitting package consists of:

(i) a finite stratification $X_{n-j}^{\mathrm{split}} = \coprod_\alpha X_\alpha$ of the ordered squarefree splitting incidence over S_{n-j};

(ii) for each α, a specified identity or Frobenius-twisted geometrically integral root cover $Y_\alpha \to X_\alpha$, with genus bound g_α and point bound

$$\#Y_\alpha(\mathbb{F}_q) \geq q + \kappa_\alpha - 2g_\alpha\sqrt{q};$$

(iii) a closed bad scheme $B_{n-j} \subset S_{n-j}$ containing the loci on which the chosen cover is not available;

(iv) branch, diagonal, old-marker, new-marker, and self-collision divisors whose total degree on Y_α is at most δ_α;

(v) bounds b and c for the pullbacks of B_{n-j} and the finite self-collision divisor to a noncontained first-polar line.

The index α names a geometric stratum together with its constant field and Frobenius twist. All degrees are geometric degrees after the displayed pullback. “Computed” means that equations for B_{n-j} and every divisor are obtained by finite contraction, elimination, factorization, and rank tests over the coefficient ring.

The construction is summarized in Figure 2. The upper branch is the level-specific component calculation; the lower branch is the general transverse argument.

$[f] \in S_n \dashrightarrow \Phi_f(U_f) \subset S_{n-1} \longrightarrow$ lower ordered splitting cover

$\searrow$ contained persistent/Lucas component,

$\nearrow$ transverse delete branch, diagonal, and markers

$\downarrow$ lift by (1)

split squarefree witness in W_f.

Fig. 2. Coherent polar induction. The contained branch is a level-specific component theorem; only the transverse branch is controlled by the general point-count argument.

Theorem V.5 (Polar-flag construction). The maps of Definition V.2 commute with field extension and the natural $\mathrm{GL}(E)$ action. They include the infinity chart displayed above. If $g \in W_{\iota_{\lambda_j} \cdots \iota_{\lambda_1} f}$, then $\lambda_1 \cdots \lambda_j g \in W_f$; the lift is squarefree exactly when g is squarefree and avoids every marker λ_i.

Proof. Contraction is defined by the perfect divided-power/symmetric-power pairing, so it commutes with base change and change of basis. Repeated application of (1) gives the membership assertion. Since the λ_i are distinct on the configuration space, their product is squarefree; the product with g is squarefree precisely when g is squarefree and none of the λ_i divides it. The coordinate formula at infinity follows by evaluating C_f on the second basis covector. □

Definition V.6 (Contained-component assertion). For a specified lower package, $\mathrm{CC}(n, j)$ is the scheme-theoretic assertion that every $f \in S_n$ for which $\Phi_f^* B_{n-j}$ or a marker-collision section vanishes identically belongs to the explicitly listed closed subscheme $\mathcal{P}_n \cup \mathcal{M}_n$. An identically zero collision differential is part of $\mathcal{M}_n$; it is never counted as a finite divisor.

This is a level-specific theorem obligation, not an assumption hidden inside the point-count argument. Sections VI and VII state the contained-component calculations used at redundancies six through nine. No assertion $\mathrm{CC}(n, j)$ is made for an arbitrary degree.

Theorem V.7 (Effective transverse polar induction). Fix a level (n, j) and a lower splitting package as in Definition V.4. Its reusable inputs are:

(i) the package's geometrically integral root covers and their triples $(g_\alpha, \delta_\alpha, \kappa_\alpha)$;
(ii) the two parameter-line degrees b, c;
(iii) the level-specific contained-component theorem $\mathrm{CC}(n, j)$.

Put

$$\mathcal{H}_\kappa(g, \delta) = \left\lfloor \left(g + \sqrt{g^2 + \delta - \kappa} \right)^2 \right\rfloor + 1$$

and assume $\delta \geq \kappa - g^2$. If

$$q \geq \max_\alpha \mathcal{H}_{\kappa_\alpha}(g_\alpha, \delta_\alpha) \quad \text{and} \quad q + 1 > b + c,$$

then every $\mathbb{F}_q$-rational Hankel system outside $\mathcal{P}_n \cup \mathcal{M}_n$ admits a rational marked polar flag whose lower squarefree witness avoids every marker. Its coherent lift is a split squarefree member of the original system. Equivalently,

$$\{\text{split-free systems in } S_n(\mathbb{F}_q)\} \subseteq (\mathcal{P}_n \cup \mathcal{M}_n)(\mathbb{F}_q).$$

Proof. For a system outside $\mathcal{P}_n \cup \mathcal{M}_n$, the assertion $\mathrm{CC}(n, j)$ makes the pulled-back bad and self-collision schemes finite of total degree at most $b + c$. Since $\#\mathbb{P}^1(\mathbb{F}_q) = q + 1$, choose a rational contraction parameter outside them. On the corresponding lower stratum the recorded point bound gives

$$q + \kappa_\alpha - 2 g_\alpha \sqrt{q} > \delta_\alpha,$$

so there is a rational point outside all deletions. It yields a split squarefree lower member avoiding every marker. Equation (1) lifts it to a split squarefree member upstairs, so the original system is shallow. Taking the contrapositive gives the displayed inclusion. □

The theorem is uniform in the numerical point-count argument and in the coherent lifting step. What must be supplied anew at each level is exactly the geometric package in (i) and the contained-component calculation in (iii); finite computation may close bounded fields but cannot replace either input.

Remark V.8 (Point-count convention). The correction κ records the exact lower bound available for the chosen model. The singular cubic fiber square of Lemma IV.6 uses $\kappa = 0$. The normalized genus-one covers in the fixed-level applications use $\kappa = 1$. Thus the redundancy-eight triple $(g, \delta, \kappa) = (1, 30, 1)$ gives the real cutoff $(1 + \sqrt{30})^2 < 42$, whose first prime power is 43; the redundancy-nine triple $(1, 36, 1)$ gives the strict cutoff $q > 49$, whose first prime power is 53.

TABLE VI
Normalized point-count thresholds used in this paper.

Application	g	δ	κ	integer lower bound	first prime power
R5 cubic fiber square	1	12	0	$q \geq 22$	23
R6 marked cover	1	18	1	$q \geq 28$	29
R7 marked cover	1	24	1	$q \geq 35$	37
R8 three-marker cover	1	30	1	$q \geq 42$	43

B. Contained-component calculations

The ordinary rank-two carrier admits a uniform contained-line calculation. This does not classify other lower bad schemes, but it closes the persistent part of every level-specific component gate.

Proposition V.9 (Contained rank-two polar lines). Let $n \geq 5$, let $f \in \mathbb{P}(\Gamma^n E)$, and let

$$A = C_f^{(2)} : \operatorname{Sym}^{n-2} E^\vee \longrightarrow \Gamma^2 E$$

be the three-row catalecticant. Assume the contraction map $E^\vee \to \Gamma^{n-1} E$ is injective, so the first-polar map is defined on all of $\mathbb{P}(E^\vee)$. This is exactly the complement of the rank-one boundary: after a change of coordinates, a nonzero kernel vector is ∂_X, and $\partial_X f = 0$ forces f to be a scalar multiple of $Y^{[n]}$. Such a point lies on the normal rational curve and is already excluded from the split-free locus. Then

$$\Phi_f(\mathbb{P}(E^\vee)) \subset \{b : \operatorname{rank} C_b^{(2)} \leq 2\} \quad \Longleftrightarrow \quad \operatorname{rank} A \leq 2$$

scheme-theoretically and in every characteristic. If $\operatorname{rank} A = 3$, the rank-drop parameters form a subscheme of $\mathbb{P}(E^\vee)$ of degree at most three.

Proof. For a nonzero $\lambda \in E^\vee$, multiplication identifies $\operatorname{Sym}^{n-3} E^\vee$ with the hyperplane

$$H_\lambda = \lambda \operatorname{Sym}^{n-3} E^\vee \subset \operatorname{Sym}^{n-2} E^\vee,$$

and functoriality gives

$$C_{\iota_\lambda f}^{(2)} = A|_{H_\lambda}.$$

The forward implication is immediate. Conversely, suppose $\operatorname{rank} A = 3$ and put $K = \ker A$, so $\dim K = n - 4$. If the lower rank is at most two, rank–nullity on the $(n-2)$-dimensional space H_λ gives

$$\dim(K \cap H_\lambda) \geq (n-2) - 2 = n - 4 = \dim K.$$

Thus $K \subset H_\lambda$. Containment for every geometric $[\lambda]$ would imply

$$K \subset \bigcap_{[\lambda] \in \mathbb{P}(E_{\bar{k}}^\vee)} \lambda \operatorname{Sym}^{n-3} E_{\bar{k}}^\vee = 0,$$

because a nonzero binary form cannot be divisible by every linear form. This contradicts $\dim K = n - 4 > 0$.

The maximal minors of $C_{\iota_\lambda f}^{(2)}$ are homogeneous cubics in λ. Vanishing at every geometric point is therefore identical vanishing, proving the scheme-theoretic assertion. In the noncontained case at least one cubic is nonzero, and the common rank-drop scheme lies in its degree-three zero divisor. □

Thus Proposition V.9 covers the entire non-rank-boundary locus on which the first-polar line is defined; the excluded noninjective locus is the rank-one normal-rational-curve locus just identified. After rank one and rational split secants are removed as shallow, the upper rank-two locus consists of the persistent tangent and conjugate-secant carriers. On a persistent fiber the nonsplit torus acts by $z \mapsto z^n$, giving T/T^n modulo inversion and Frobenius, while the tangent unipotent cocycle is $z \mapsto z + nu$.

Modular carriers are equally explicit. If $\mathcal{N}$ is a nucleus in the lower divided-power module, the complete space whose first-polar line lies in $\mathcal{N}$ is

$$\mathcal{M}_n(\mathcal{N}) = \mathbb{P} \ker \left(\Gamma^n E \longrightarrow \operatorname{Hom}(E^\vee, \Gamma^{n-1} E / \widetilde{\mathcal{N}}) \right).$$

Lucas' theorem computes the nucleus coordinates, and this displayed kernel computes every coherent lift without an ambient syndrome census.

Remark V.10 (Scope of the contained calculations). The modular contraction kernel is an exact finite linear-algebra construction. Proposition V.9 supplies the ordinary persistent component uniformly, but other lower bad schemes and marker collisions still require level-specific calculations. Neither the proposition nor Theorem V.7 classifies those additional components.

VI. Redundancies six and seven: complete and gated results

A. Redundancy six

For

$$f = (a_0 : \cdots : a_5) \in \mathbb{P}(\Gamma^5 E), \qquad W_f = \ker \begin{pmatrix} a_0 & a_1 & a_2 & a_3 & a_4 \\ a_1 & a_2 & a_3 & a_4 & a_5 \end{pmatrix},$$

the system W_f is a net of quartics off the quintic normal rational curve. It is split-free exactly when it has no completely split squarefree quartic.

Proposition VI.1 (Persistent net and orbit law). If $\gcd(W_f) = Q$ has degree two, then

$$W_f = Q \operatorname{Sym}^2 E^\vee.$$

The deep persistent locus has size $q(q+1)^2/2$. On the tangent fiber the unipotent coordinate is $x \mapsto x + 5u$; on the irreducible quadratic fiber the norm-one torus acts through $z \mapsto z^5$, with inversion and coefficient Frobenius acting on T/T^5.

Proof. For fixed Q, the syndromes annihilating $Q \operatorname{Sym}^3 E^\vee$ form a projective line. If Q is irreducible, all $q+1$ points remain; if $Q = \lambda^2$, one endpoint has dependent Hankel rows, leaving q points. Hence

$$q(q+1) + \frac{q(q-1)}{2}(q+1) = \frac{q(q+1)^2}{2}.$$

Normalize a square to U^2. The nondegenerate fiber is $[e_4 + xe_5]$, and direct degree-five substitution gives $x \mapsto x + 5u$. For an irreducible Q, its stabilizer identifies the $q+1$-point fiber with $T = \{z \in \mathbb{F}_{q^2}^\times : z^{q+1} = 1\}$ and acts by fifth powers; the normalizer inverts z, while coefficient Frobenius multiplies T/T^5 by the characteristic. This proves the stated orbit law and the characteristic-five tangent splitting. □

Proposition VI.2 (First-polar line and secant degree). For $r \in \mathbb{F}_q$ put

$$b(r) = (a_1 - ra_0, \ldots, a_5 - ra_4), \qquad b(\infty) = (a_0, \ldots, a_4).$$

Then

$$(T - rU)g \in W_f \iff g \in W_{b(r)}.$$

If W_f has trivial gcd, its polar line meets the lower secant cubic in degree at most three.

Proof. The first equivalence follows by expanding the two Hankel equations of $(T - rU)g$. Let

$$C_f = \begin{pmatrix} a_0 & a_1 & a_2 & a_3 \\ a_1 & a_2 & a_3 & a_4 \\ a_2 & a_3 & a_4 & a_5 \end{pmatrix}.$$

The lower catalecticant equals

$$C_f \begin{pmatrix} -r & 0 & 0 \\ 1 & -r & 0 \\ 0 & 1 & -r \\ 0 & 0 & 1 \end{pmatrix}.$$

Cauchy–Binet gives the secant determinant

$$D_f(r) = -\Delta_{012} r^3 + \Delta_{013} r^2 - \Delta_{023} r + \Delta_{123}.$$

It vanishes identically exactly when $\operatorname{rank} C_f \le 2$, equivalently when W_f has quadratic gcd. In the trivial-gcd case it therefore has at most three zeros. □

Proposition VI.3 (Cyclic and collision degrees). For a trivial-gcd net:

(i) in characteristic different from two and three, the lower tame cyclic carrier is a binomially rescaled Veronese surface of degree four and contains no polar line;
(ii) the pointed ramification condition has degree at most six and does not vanish identically;
(iii) an S_3 lower pencil with the marked root deleted has total deletion degree at most eighteen.

Proof. In syndrome coordinates the tame cyclic surface is

$$\left\{ \operatorname{diag}(1, 4, 6, 4, 1)^{-1} [Q^2] : [Q] \in \mathbb{P}(\operatorname{Sym}^2 E^\vee) \right\}.$$

When 2 and 3 are invertible this is the Veronese surface, of degree four, and contains no line. Characteristic three has a different wild closure, treated in Proposition VI.4. For a basis F_0, F_1, F_2 of the $g^2_4 = W_f$, pointed collision is the rank-one condition

$$\operatorname{rank}\begin{pmatrix} F_0(r) & F_1(r) & F_2(r) \\ F_0'(r) & F_1'(r) & F_2'(r) \end{pmatrix} \leq 1.$$

Its Wronskian divisor has degree six. In characteristic at least five it cannot vanish identically because the order sequence is classical. In characteristic three an identically zero first-derivative minor would make the net a pullback from a series of degree at most one, hence of vector dimension at most two. In characteristic two the only dimension-three equality case is the pure-square space $\langle 1, t^2, t^4\rangle$. If two consecutive Hankel rows had this common kernel, comparison of the overlapping coefficients would force $a_0 = \cdots = a_5 = 0$. Thus some derivative minor is nonzero in both modular characteristics. If instead the derivative row is everywhere proportional to the value row, the corresponding rational map is inseparable; after cancelling its pencil gcd, the same dimension argument places it on the fixed-factor boundary. Hence the ramification scheme is a genuine divisor of degree at most six.

Finally the R5 off-diagonal curve has the original deletion degree twelve. The unique cubic through a specified marker contributes at most six ordered pairs, giving $\delta = 18$ and

$$q - 2\sqrt{q} > 18.$$

Its first prime-power solution is 29. □

Proposition VI.4 (Modular contained components). In characteristic three the lower wild carrier is the degree-four cone

$$\operatorname{Join}(e_2, C_4),$$

and no polar line of a trivial-gcd net is contained in it. In characteristic two the cyclic carrier is the plane

$$\Pi_{\mathrm{cyc}} = \mathbb{P}\langle e_1, e_2, e_3\rangle,$$

whose unique contained polar-line component is

$$\mathcal{N}_3 = \mathbb{P}\langle e_2, e_3\rangle.$$

Proof. Every line on the wild cone is a ruling. By equivariance it suffices to compare the two consecutive rows on one ruling; in the frame $\langle e_2, e_4\rangle$ the conditions

$$(a_0, \ldots, a_4),\ (a_1, \ldots, a_5) \in \langle e_2, e_4\rangle$$

force the overlapping coordinates successively to vanish. Undoing the frame gives $f = \mu\nu(t)$, including $f = \mu e_5$ at infinity. These are fixed-factor/rank-boundary syndromes, so no trivial-gcd polar line is contained in the wild cone.

In characteristic two, direct substitution gives

$$g \cdot e_2 = [0 : \delta\gamma : \alpha\delta + \beta\gamma : \alpha\beta : 0],$$

whose closure is Π_{cyc}. The consecutive-row overlap is now literal: the first row lies in the plane exactly when $a_0 = a_4 = 0$, and the second exactly when $a_1 = a_5 = 0$. Hence a polar line lies in the plane exactly when

$$a_0 = a_1 = a_4 = a_5 = 0,$$

namely when $f \in \mathbb{P}\langle e_2, e_3\rangle$. This proves both existence and uniqueness of the contained binary component. □

Proposition VI.5 (Binary nucleus arithmetic). The scheme $\mathcal{N}_3(\mathbb{F}_q)$ is one projective-semilinear orbit of $q + 1$ points. It is split-free over $q = 2^m$ exactly when m is odd.

Proof. For the representative e_3,

$$W_{e_3} = \langle 1, t, t^4\rangle.$$

The root differences of $A + Bt + Ct^4$ lie in the kernel of $u \mapsto u^4 + (B/C)u$. Four distinct rational roots exist exactly when all roots of $u^3 = B/C$ are rational, equivalently when $3 \mid q - 1$, which for $q = 2^m$ means m is even. The stabilizer of e_3 is the Borel of order $q(q-1)$, so the orbit has $q + 1$ points; its lower-unipotent transforms and endpoint are precisely $\mathcal{N}_3(\mathbb{F}_q)$. □

The covering-radius gate is complete: Seroussi–Roth gives $\rho(\mathrm{PRS}(q-5)) = 5$ for $q \geq 9$, while Certificate R6 checks it directly at $q = 7, 8$.

Theorem VI.6 (Redundancy six). For every prime power $q \geq 7$, the deep syndromes of $\mathrm{PRS}(q-5)$ are the persistent stratum of size $q(q+1)^2/2$, together with:

(i) respectively $18, 11, 4, 2, 1$ exceptional $\mathrm{PGL}_2(q)$ orbits at $q = 7, 8, 9, 11, 13$;

(ii) one characteristic-two nucleus orbit of size $q+1$ when $q=2^m$ and odd $m \geq 5$.

There are no other exceptional orbits. The complete persistent orbit law is T/T^5 modulo inversion and Frobenius, with a tangent fixed/nonzero split exactly in characteristic five.

Proof. Propositions VI.1–VI.5 classify the common-factor and contained components and exclude a trivial-gcd exception in the conceptual ranges: characteristic at least five from $q \geq 29$, characteristic three from its next unchecked field $q = 81$, and characteristic two outside $\mathcal{N}_3$ from $q \geq 32$. Certificate R6 exhausts

$$q \in \{7, 8, 9, 11, 13, 16, 17, 19, 23, 25, 27\},$$

finds exceptional projective-orbit counts $18, 11, 4, 2, 1$ only at the five displayed fields, and no further orbit at the remaining fields. Together with the separate radius gate, this covers every prime power $q \geq 7$. □

The five small tables are exhaustive computations. Their semilinear orbit counts are $18, 5, 2, 2, 1$; a collision-energy invariant plus, in odd characteristic, the quintic root-divisor type separates the classes up to precisely Frobenius fusion. Certificate R6 supplies a canonical representative, stabilizer, member histogram, and Frobenius link for every projective orbit, while Certificate R6-NF supplies the collision-energy and root-divisor fingerprint. Orbit counts alone are not used as identifiers.

B. Redundancy seven

For

$$f = (a_0 : \ldots : a_6) \in \mathbb{P}(\Gamma^6 E), \qquad W_f = \ker \begin{pmatrix} a_0 & a_1 & a_2 & a_3 & a_4 & a_5 \\ a_1 & a_2 & a_3 & a_4 & a_5 & a_6 \end{pmatrix},$$

the system W_f is a four-dimensional web of quintics. For an affine marker r the contractions are

$$b(r) = (a_1 - ra_0, \ldots, a_6 - ra_5).$$

Coefficient collection gives

$$(t-r)g \in W_f \quad \Longleftrightarrow \quad g \in W_{b(r)};$$

the lift is squarefree exactly when $g(r) \neq 0$.

Proposition VI.7 (Two-marker lower cover). Let x be a forbidden root on a trivial-gcd quartic net. On the generic S_3 bottom fiber, the diagonal and branch divisors and the two marked-root fibers have total degree at most

$$12 + 6 + 6 = 24.$$

Hence $q - 2\sqrt{q} > 24$, and in particular every prime power $q \geq 37$, produces a split lower quartic avoiding x.

Proof. The R5 off-diagonal curve is geometrically integral of arithmetic genus one and has base deletion degree twelve. Each marker belongs to a unique cubic fiber, which contains at most six ordered pairs. Delete those two fibers and apply the $\kappa = 0$ bound. □

Proposition VI.8 (Exact-gcd-one avoidance). Suppose the lower net is $\ell_x U$, where $U = \ker \lambda \subset \operatorname{Sym}^3 E^\vee$ has dimension three and $x \neq r$. For $q \geq 16$ it contains a split cubic avoiding both x and r.

Proof. Put $S = \mathbb{P}^1(\mathbb{F}_q) \setminus \{x, r\}$. For ordered distinct $u, v \in S$, the map $w \mapsto \lambda(\ell_u \ell_v \ell_w)$ is linear. If it vanishes identically, choose any remaining w; otherwise it has one rational zero. The equations $w = x, r, u, v$ have respective bidegrees $(1,1), (1,1), (2,1), (1,2)$ in (u,v). None is identically zero: the first two would add a common factor, and the last two are excluded because double-root cubics span $\operatorname{Sym}^3 E^\vee$. A nonzero bidegree-$(a,b)$ divisor on $\mathbb{P}^1 \times \mathbb{P}^1$ has at most $(a+b)(q+1)$ rational points. Thus the bad pairs number at most $10(q+1)$, whereas $(q-1)(q-2) > 10(q+1)$ for $q \geq 16$. □

Proposition VI.9 (Marked self-collision). After fixed factors are removed, the self-collision divisor of the moving g^3_5 has degree at most eight.

Proof. It is the ramification divisor of a separable g^3_5, whose degree is $(3+1)(5-3) = 8$. If the series were inseparable, it would factor through Frobenius and lie in the p-th-power subspace of quintics; that subspace has dimension at most three, smaller than the four-dimensional Hankel series. □

The characteristic-two nucleus line at redundancy six has exactly one coherent lift, the central point e_3, with

$$W_{e_3} = \langle 1, t, t^4, t^5 \rangle.$$

It is deep for odd m; for even m, the homogenization of $t^4 + t$ has the five roots $\mathbb{P}^1(\mathbb{F}_4)$.

Proposition VI.10 (Central binary lift). The complete inverse image of the R6 nucleus line under consecutive sextic contraction is $[e_3]$. It is split-free over $\mathbb{F}_{2^m}$ exactly when m is odd.

Proof. Putting both consecutive rows in $\mathbb{P}\langle e_2, e_3\rangle$ gives $a_0 = a_1 = a_2 = a_4 = a_5 = a_6 = 0$. Thus only e_3 remains, with $W_{e_3} = \langle 1, t, t^4, t^5\rangle$. For odd m, contraction at a root of a hypothetical split quintic would give a split quartic on the R6 nucleus line, contradicting Proposition VI.5. For even m, $\mathbb{F}_4 \subset \mathbb{F}_q$ and the homogenization of $t^4 + t$ has its four affine $\mathbb{F}_4$ roots and the simple root at infinity. □

Corollary VI.11 (Degree-six contained component). The assertion $\mathrm{CC}(6, 1)$ for the lower rank-two carrier holds in every characteristic. Its contained sextics have upper catalecticant rank at most two; after the shallow rank-one and rational secant loci are removed, they are exactly the persistent tangent and conjugate-secant families.

Proof. Apply Proposition V.9 with $n = 6$. The three rational types of the quadratic recurrence are split squarefree, repeated rational, and irreducible. They give respectively the shallow rational secant, tangent, and conjugate-secant loci. The remaining lower binary nucleus line has the separate complete lift computed in Proposition VI.10. □

Theorem VI.12 (Redundancy-seven classification). Certificate R7 unconditionally classifies every prime power $q \leq 32$ in its displayed domain. For every prime power $q \geq 13$, the split-free syndromes at redundancy seven are the persistent stratum, together with the central nucleus point when $q = 2^m$ and m is odd. At $q = 7, 8, 9, 11$ the additional split-free orbit-size profiles are:

$$\begin{array}{r|l} 7 & 56,\ 84^5,\ 112^2,\ 168^{45},\ 336^{141} \\ 8 & 63,\ 72,\ 84^3,\ 168^4,\ 252^{24},\ 504^{86} \\ 9 & 180^3,\ 240^6,\ 360^{18},\ 720^{27} \\ 11 & 264^2,\ 440,\ 660^2. \end{array}$$

This is the complete all-field split-free classification. Since Seroussi–Roth gives $\rho(\mathrm{PRS}(q-6)) = 6$ for every prime power $q \geq 11$, it is also the complete deep-hole classification in that range. At $q = 7, 8, 9$ the table is not promoted to a code deep-hole classification without a separate covering-radius calculation. The persistent orbit law is T/T^6 modulo inversion and Frobenius, with tangent splitting in characteristics two and three.

Proof. For $q \geq 37$, Propositions VI.7–VI.9 give the transverse witness after at most three catalecticant intersections, eight self-collisions, and, in characteristic two, one intersection with the nucleus line. Corollary VI.11 leaves exactly the persistent locus and Proposition VI.10. Certificate R7 covers

$$q \in \{7, 8, 9, 11, 13, 16, 17, 19, 23, 25, 27, 29, 31, 32\}$$

by an independent five-secant replay. The radius theorem promotes the split-free list only from $q \geq 11$. □

For the four finite rows Certificate R7 contains the canonical representative, stabilizer, persistent/central flags, and Frobenius link of every orbit. The displayed size profile is only a compact summary; repeated sizes are distinguished by those canonical records.

The $q = 19$ lower net $\langle 1, t^3, t^4\rangle$ has six split members, all containing the marked point at infinity, yet no coherent sextic polar line lies in its orbit. This is the concrete reason Theorem V.7 is formulated for parameterized flags rather than a set of bad lower fibers.

TABLE VII

Field-range closure ledger. "Certificate" means a complete finite classification with canonical representatives, not merely an orbit count.

Result	prime powers	closure mechanism
R5	$7, 8, 9, 11, 13, 16, 17, 19$	direct census and independent replay
	$q \geq 23$	cubic-cover geometry, Lemma IV.6
R6	$7, 8, 9, 11, 13, 16$	direct syndrome/radius census
	$17, 19, 23, 25, 27$	finite structural bridge certificate
	$q \geq 29$	marked-cover point bound and contained components
R7	$7, 8, 9, 11$	direct split-free census
	$13, 16, 17, 19, 23, 25, 27, 29, 31, 32$	coherent-polar finite bridge certificate
	$q \geq 37$	marked-cover point bound and contained components

The R7 rows classify split-free syndromes. Conversion to code deep-hole directions uses the separate covering-radius gate and is made only for $q \geq 11$. The public certificate schema and exact replay boundaries are given in the supplement; the table does not promote orbit-size profiles to identifiers.

VII. A fixed-level high-field application

A. Redundancy eight

Proposition VII.1 (Generic three-marker bottom cover). After three contractions, the generic bottom object is the R5 off-diagonal $(2,2)$ curve. The diagonal and branch deletion costs twelve and three marked roots cost eighteen, so $\delta = 30$. The inequality

$$q + 1 - 2\sqrt{q} > 30$$

holds at every prime power $q \geq 43$. The first-polar line has transverse/collision budget

$$3 + 1 + 10 = 14,$$

where ten is the ramification degree of a moving g_6^4.

Proof. Markers do not change the bottom curve or its geometric monodromy; each removes at most the six ordered pairs in its unique cubic fiber. The point bound is the $\kappa = 1$ row of Table VI. A noncontained line meets the lower rank-two catalecticant carrier in degree at most three, and the lower binary central point in at most one parameter. After fixed factors are removed, self-collision is ramification of a separable g_6^4, of degree $(4+1)(6-4) = 10$. Separability follows because the p-th-power subspace has dimension smaller than the five-dimensional series. □

Definition VII.2 (Three-marker lower-package gate). The assertion $\mathrm{LP}(6,1)$ says that the pointed R7 splitting incidence, after the persistent rank-two carrier, the binary central point, and the marked-collision divisor are removed, has the geometrically integral identity twists and deletion bounds used in Proposition VII.1. It includes equations and monodromy checks for the recursive cyclic, wild, inseparable, gcd, and branch strata; a finite-field regression is not a substitute for those geometric data.

The ordered cover retains the marker labels needed for monodromy, but the contraction itself descends to the unordered marker cubic. Indeed, for affine markers r, s, t,

$$a_i^{\langle r,s,t\rangle} = a_{i+3} - (r+s+t)a_{i+2} + (rs+rt+st)a_{i+1} - rst\, a_i. \tag{2}$$

Thus it depends only on the elementary symmetric functions of the markers, equivalently on the monic cubic $T^3 - (r+s+t)T^2 + (rs+rt+st)T - rst$. This separates the symmetric contraction parameter from the ordered-root cover used below.

Lemma VII.3 (Bottom strata). For the two-step contraction of a pointed sextic, the lower rank, gcd-one, cyclic, wild, inseparable, and branch strata are exactly the closed schemes displayed below. The reversed homogeneous chart introduces no additional stratum.

Proof. For a pointed sextic $f = (a_0, \ldots, a_6)$ with old marker x, contract successively:

$$b_i(r) = a_{i+1} - ra_i, \qquad c_i(r,s) = b_{i+1}(r) - sb_i(r).$$

For $c = (c_0, \ldots, c_4)$ put

$$H_c = \begin{pmatrix} c_0 & c_1 & c_2 & c_3 \\ c_1 & c_2 & c_3 & c_4 \end{pmatrix}.$$

On the rank-two-kernel chart, if p_0, p_1 span $\ker H_c$, the bottom ordered-root curve is

$$G_c(u,v) = \frac{p_0(u)p_1(v) - p_1(u)p_0(v)}{[u,v]} = 0. \tag{3}$$

The rank-at-most-two secant carrier, including the gcd-two and rank-drop boundary strata, is

$$D(c) = \det \begin{pmatrix} c_0 & c_1 & c_2 \\ c_1 & c_2 & c_3 \\ c_2 & c_3 & c_4 \end{pmatrix} = 0. \tag{4}$$

For an exact common root z, the gcd-one incidence is cut out by

$$\operatorname{rank} \begin{pmatrix} H_c & & & \\ 1 & z & z^2 & z^3 \end{pmatrix} \leq 2. \tag{5}$$

with the evident homogeneous row at infinity. After substitution, (4) has degree at most three in s, while each 3-minor in (5) has degree at most two for fixed z.

The remaining reducible-monodromy carriers are explicit. Put $z_i = \binom{4}{i} c_i$. Outside characteristics two and three the cyclic closure is

$$[u : v : w] \longmapsto [u^2 : 2uv : v^2 + 2uw : 2vw : w^2], \tag{6}$$

a degree-four projected Veronese surface containing no line. In characteristic three it is the degree-four cone $\bowtie(e_2, C_4)$; its only lines are rulings, and the consecutive-row overlap excludes every ruling as a polar line. In characteristic two it is

$$\Pi_{\text{cyc}} = \mathbb{P}\langle e_1, e_2, e_3\rangle, \qquad c_0 = c_4 = 0. \tag{7}$$

whose unique contained polar-line component is $\mathcal{N}_3 = \mathbb{P}\langle e_2, e_3\rangle$. The inseparable degeneration is the derivative-minor locus of p_0, p_1 and introduces no further stratum. Outside characteristics two and three a nonconstant degree-three map cannot factor through Frobenius. In characteristic three an inseparable cubic pencil is projectively $\langle X^3, Y^3\rangle$; the two Hankel rows then force $c_0 = c_1 = c_3 = c_4 = 0$, the vertex e_2 of the wild cone. In characteristic two, after cancelling a nonzero pencil gcd, an inseparable ratio has degree at most two and is a rational function of t^2. Homogenizing its numerator and denominator to cubics supplies the same linear factor to both, so it belongs to the exact-gcd-one boundary. Finally, the branch equation on this chart is $p_0'p_1 - p_0p_1' = 0$. These carrier and branch loci are the scheme-theoretic images of the displayed incidences (equivalently their Fitting-minor ideals); the reversed homogeneous chart supplies infinity. Thus these are closed strata over the coefficient ring, not a classification only of their rational points. □

Lemma VII.4 (Bottom monodromy and deletion). Off the strata of Lemma VII.3, the bottom ordered-root curve is geometrically integral of arithmetic genus one and has deletion degree at most 30. On the exact-gcd-one stratum, the residual deck graph supplies a squarefree pair avoiding the three markers for $q \geq 43$.

Proof. Off (4)–(7), the separable cubic map is noncyclic, so its geometric monodromy is S_3. Transitivity on ordered pairs of distinct roots makes (3) geometrically integral. It has arithmetic genus one. The diagonal costs four, the degree-four different costs at most eight off-diagonal ordered pairs, and each of the three marker fibers x, r, s costs at most six. Thus

$$\delta \leq 4 + 8 + 3 \cdot 6 = 30. \tag{8}$$

Hasse–Weil supplies a surviving rational point for every $q \geq 43$.

It remains to show that the strata removed above do not hide another component. On an exact-gcd-one bottom stratum write the cubic pencil as $\ell_z Q$, with Q a basepoint-free pencil of quadratics. In the separable case its deck involution ι is rational, so its graph has $q + 1$ rational points. At most two are fixed and at most two ordered graph points involve each of z, x, r, s. A point remains for $q \geq 43$, giving three distinct roots outside the markers. The inseparable degree-two case is on the rank/boundary carrier. □

Lemma VII.5 (Two-marker selection). Fix two old markers x, r. Outside the declared contained strata, the bad divisor on the internal s-line has degree at most 19. For $q \geq 43$ one can choose s so that the lower package supplies a split squarefree quartic avoiding x and r.

Proof. The complete divisor table on the s-line is

failure	defining mechanism	degree
rank at most two	$D(c(s))$	3
cyclic/wild monodromy	pullback of (6)–(7)	4
new gcd root s	$\mathrm{Ram}(g_d^2),\ d \leq 4$	6
gcd root x or r	one nonzero minor in (5)	$2+2$
marker collision	$s = x,\ s = r$	2

In characteristic two the cyclic entry has degree one, so degree four is a uniform bound. The total is

$$3 + 4 + 6 + 2 + 2 + 2 = 19 < q + 1. \tag{9}$$

The rank pullback cannot be the whole line: Proposition VI.2 identifies that containment exactly with quadratic gcd of the original net. The only possible contained cyclic line is the declared binary nucleus. For a fixed marker, not all minors in (5) can vanish identically without giving the original quartic net that fixed factor: otherwise the hyperplanes $W \cap \ker(\mathrm{ev}_s)$ and $W \cap \ker(\mathrm{ev}_x)$ agree for dense s, making the evaluation map constant. Choose one nonzero quadratic minor. For self-collision, take a generic pencil projection of the moving g_d^2, $d \leq 4$. It is separable: in characteristic two the only dimension-three equality case is the pure-square space $\langle 1, t^2, t^4\rangle$, which the consecutive Hankel overlap equations do not realize; all other p-power spaces have smaller dimension. Riemann–Hurwitz gives degree at most $2d - 2 \leq 6$, and every self-collision maps into this ramification divisor. Choosing s outside this divisor and applying the preceding gcd-zero or gcd-one bottom analysis gives a split squarefree quartic avoiding the two old markers. This proves the two-old-marker lemma. □

Lemma VII.6 (Outer marker selection). For a pointed sextic outside the persistent and modular contained strata, the bad divisor on the outer first-marker line has degree at most 15. A surviving outer marker lifts the witness of Lemma VII.5 to a split squarefree quintic avoiding the original marker.

Proof. The lower rank-two pullback costs three parameters and the binary nucleus costs one. If a contracted net has exact gcd ℓ_z, the condition $z = r$ maps into the ramification of a generic separable pencil projection of the moving g^3_d, $d \leq 5$. Riemann–Hurwitz bounds it by $2d - 2 \leq 8$. The condition $z = x$ is again (5), now with the quartic evaluation row, and costs at most two because not all of its quadratic minors vanish off the fixed-marker boundary, by the same evaluation-hyperplane argument. Together with $r = x$, the internal divisor has degree

$$3 + 1 + 8 + 2 + 1 = 15. \tag{10}$$

If the contracted net has trivial gcd, the two-old-marker lemma applies. If it has exact gcd with $z \notin \{x, r\}$, the ordered-pair argument has three $(1, 1)$ and two $(2, 1)/(1, 2)$ bad equations, at most $12(q + 1) < (q - 2)(q - 3)$. The cases $z = x, r$ are precisely the excluded quadratic and ramification divisors. Lifting first by ℓ_s and then by ℓ_r gives a split squarefree quintic avoiding x. ☐

Proposition VII.7 (Pointed lower package). The assertion $\mathrm{LP}(6, 1)$ holds. On the outer first-marker line its exceptional divisor has degree at most fifteen; on the recursive second-marker line the corresponding divisor has degree at most nineteen. Its only positive-genus identity twist is the geometrically integral off-diagonal $(2, 2)$ curve of genus at most one with deletion degree 30.

Proof. Lemmas VII.3 and VII.4 identify the bottom strata and the unique positive-genus identity twist. Lemmas VII.5 and VII.6 give the two parameter-line budgets and the coherent squarefree lift. For auditability, the complete exhaustion is summarized here.

Stratum	Defining equation	Degree	Contained case	Transverse conclusion
rank at most two	(4)	3 in the marker	quadratic-gcd persistent carrier	finite intersection of degree at most 3
cyclic or wild	(6) or (7)	at most 4	only the binary nucleus line	finite intersection of degree at most 4
exact gcd one	minors in (5)	2 per fixed marker	fixed-factor boundary	the residual deck graph supplies a squarefree pair
inseparable	derivative minors of p_0, p_1	absorbed above	wild vertex, rank drop, or binary nucleus	no additional stratum
geometric S_3	(3)	bidegree $(2, 2)$	only the preceding carriers	integral genus-one cover with deletion degree 30
marker collision	diagonals and ramification	1 per diagonal; at most 6 for self-collision	declared collision boundary	finite divisor on the marker line

The gcd has degree zero, one, or at least two, and a separable gcd-zero cubic cover has monodromy S_3 or C_3; hence the displayed list is exhaustive. No certificate or finite-field regression is used for this integrality assertion. Notice also that the outer and recursive parameter budgets 15 and 19 are already strictly below $q + 1$ for $q \geq 19$; the threshold 43 is imposed solely by the genus-one deletion inequality (8), not by a hidden parameter-choice bound. ☐

Proposition VII.8 (Modular lifts at degree seven). The consecutive-row Lucas calculation leaves no nonzero lift in characteristic two, the line $\mathbb{P}\langle e_2, e_5\rangle$ in characteristic three, and the line $\mathbb{P}\langle e_3, e_4\rangle$ in characteristic five. The latter two lines are shallow in the range $q \geq 43$.

Proof. Lucas' criterion applied to the consecutive binomial rows gives the three stated supports; the exact row reductions are replayed by Certificate R8. In characteristic five, the homogenization of $t^5 - t$ supplies six distinct projective roots and satisfies the two required middle-coefficient equations throughout the lift.

In characteristic three, $\mathrm{PGL}_2(q)$ is transitive on the rational points of $\mathbb{P}\langle e_2, e_5\rangle$, so it suffices to treat e_2. Choose nonzero a, b, c with no two equal up to sign and $a^2 + b^2 + c^2 = 0$. The nonsingular conic has $q + 1$ points; deleting the three coordinate and six equality/antiequality sections costs at most eighteen, so a choice exists for $q \geq 27$. The reciprocal pairs $\pm a^{-1}, \pm b^{-1}, \pm c^{-1}$ give six distinct roots and the two required elementary-symmetric coefficients vanish. ☐

Proposition VII.9 (Degree-seven contained components). The assertion $\mathrm{CC}(7, 1)$ for the structural components of Definition VII.2 holds in every characteristic. The rank-two component lifts to the upper rank-two carrier; the lower binary central point has no nonzero coherent lift; and the marked-collision scheme is never the whole polar line. The only additional nucleus lifts are

$$\mathbb{P}\langle e_2, e_5\rangle \quad \text{in characteristic three,} \qquad \mathbb{P}\langle e_3, e_4\rangle \quad \text{in characteristic five,}$$

and both are shallow in the range $q \geq 43$.

Proof. The rank-two assertion is Proposition V.9 with $n = 7$. For the binary central point e_3, requiring both consecutive septic rows to be supported at its single coordinate permits a_3 in the first row and a_4 in the second; their overlap is zero. Thus its projective coherent lift is empty. The full nucleus-support overlap is exactly Proposition VII.8, which also supplies the shallow witnesses in characteristics three and five.

It remains to exclude an identically colliding series. Remove the fixed divisor from the five-dimensional W_f and let the moving degree be $d \leq 6$. If its projective map were inseparable in characteristic p, it would factor through Frobenius, so its section space would have dimension at most

$$\lfloor d/p \rfloor + 1 \leq 4,$$

contrary to $\dim W_f = 5$. A generic pencil projection is therefore a separable map of degree at most six. Every marked self-collision is a ramification point of that projection, so Riemann–Hurwitz bounds the collision divisor by $2d - 2 \leq 10$. It is finite and cannot be an additional contained component. □

Remark VII.10 (Exact remaining R8 gate). Proposition VII.9 closes the contained-component problem, while Proposition VII.7 closes the pointed lower package. The certificate checks the nucleus overlaps, witnesses, and numerical budgets; the geometric integrality and recursive carrier equations are proved separately above.

Theorem VII.11 (Redundancy eight). For every prime power $q \geq 43$, the deep syndromes of $\mathrm{PRS}(q-7)$ are exactly the persistent tangent and conjugate-secant families, of total size $q(q+1)^2/2$. Their orbit law is T/T^7 modulo inversion and Frobenius. The tangent family splits exactly in characteristic seven.

Proof. Outside the contained locus, Proposition VII.1 chooses a good first contraction, and Proposition VII.7 supplies a three-marker witness. Propositions VII.8 and VII.9 eliminate every nonpersistent contained lift. The imported covering-radius theorem promotes the resulting split-free classification to deep holes. □

This theorem makes no assertion about additional deep orbits for $q < 43$.

VIII. Verification and formal boundary

The conceptual theorems above are accompanied by deterministic evidence bundles. Table VIII records which claims use exhaustive computation. Each public label resolves, through the supplement schema, to its generator, certificate, replay where stated, and SHA-256 manifest. Table I gives the corresponding claim-level mathematical dependencies.

TABLE VIII: Public verification map. Exact filenames, commands, hashes, domains, and stop conditions belong to the electronic supplement.

Public label	Verified domain	Replay and trust boundary
Certificate R5	nineteen finite fields	independent pointwise and orbitwise replay; canonical representatives and exhaustion records
Certificate R6	direct scan through $q = 16$; finite bridge below 29	independent syndrome replay and separate radius gate
Certificate R6-NF	small exceptional normal forms	deterministic normal-form reconstruction; not a second implementation
Certificate R7	$q = 7, 8, 9, 11$ and finite bridge below 37	independent five-secant/orbit checks; split-free and radius flags remain separate
Certificate R8	fixed-level characteristic cases and bounds	independent algebra/nucleus replay; not an ambient census

The public labels are defined by supplement/CERTIFICATE-SCHEMA.md. The paper-local bundle, literal replay commands, toolchain locks, hashes, byte counts, and working directories are recorded in supplement/REPRODUCING.md, supplement/EVIDENCE-MANIFEST.json, and supplement/RELEASE-MANIFEST.md. The single entry point python3 supplement/verify.py checks the bundle manifest, classification records and their expected hashes, and the local PDF and supplement-artifact rows printed in the release manifest; its –replay option runs the paper-local Python replays.

The Lean development has one paper-facing aggregate import gate. It checks the common Hankel and coding interfaces; redundancy-five Hankel algebra, projective scaling, family/count arithmetic, and transcribed finite-table arithmetic; coherent polar contraction and the redundancy-six/seven conditional synthesis interfaces; the redundancy-eight marker algebra, budgets, threshold arithmetic, and conditional synthesis.

The formal development does not prove the concrete projective Hankel dictionaries, geometric component classifications, Hasse–Weil existence after deletion, covering-radius theorems, genuine $\mathrm{PGL}_2 / \mathrm{P\Gamma L}_2$ actions, or the semantics and exhaustion of the external finite records. These remain theorem hypotheses or structure fields. The aggregate axiom audit reports only propext, Classical.choice, and Quot.sound; there is no project-local axiom, generated evaluator, native decision procedure, or external oracle in the imported closure. The exact declaration for every manuscript theorem, its missing formal content, the aggregate gate, and the reproduction command are listed in supplement/LEAN-STATEMENTS.md.

Data, code, and disclosure

The electronic supplement accompanying this manuscript contains the source, canonical classification representatives and stabilizers, certificate schemas, generators, independent replays where available, expected hashes, toolchain locks, and a top-level verification command. Section VIII assigns each computational or formal claim its exact trust route.

IX. Scope and open problems

The fixed-level theorems and the contained/transverse fork point to one stable picture. Their point-count thresholds are compatible with a quadratic uniform scale in the redundancy, while every surviving infinite family comes from one of the two component mechanisms isolated by the polar flag.

Conjecture IX.1 (Stable polar classification). There is an absolute constant C such that, at redundancy r, every split-free projective syndrome over $\mathbb{F}_q$ with $q \geq Cr^2$ lies either in a persistent rank-two carrier or in a nonshallow component of the level-r Lucas contraction kernel. Equivalently, transverse arithmetic exceptions disappear above a quadratic field threshold. In every characteristic for which that Lucas kernel has no nonshallow component, the persistent tangent and conjugate-secant families are the complete high-field list.

The Lucas clause is forced by the binary nucleus families at redundancies six and seven, which persist over infinitely many fields. The conjecture predicts that rank-two persistence and digit-controlled contraction kernels exhaust the stable locus, with the coherent polar flag eliminating everything transverse.

The present results establish the first levels of this picture but do not constitute an arbitrary-redundancy classification.

1) Redundancy five is complete for every $q \geq 7$.
2) Redundancy six is a complete all-field deep-hole classification. Redundancy seven has a complete all-field split-free classification; its $q = 7, 8, 9$ covering-radius gate remains separate.
3) Redundancy eight is complete for $q \geq 43$. No bounded-field completion is inferred.
4) The effective polar-induction theorem requires a new level-specific monodromy/component calculation at each redundancy. A field census cannot replace this input.

The exact bounded-field census gap is finite but presently uncomputed. For redundancy eight it consists of

$$q \in \{7, 8, 9, 11, 13, 16, 17, 19, 23, 25, 27, 29, 31, 32, 37, 41\};$$

the existing R8 certificate does not enumerate these ambient syndrome spaces. A direct scan at the upper endpoint would visit

$$\#\mathbb{P}^7(\mathbb{F}_{41}) = 199623130728$$

directions. Even quotienting by the maximum possible $\mathrm{PGL}_2(q)$ orbit size leaves at least 2898130 orbit representatives. Thus a bounded completion requires a new canonical-orbit enumerator, measured resource bounds, and an independent replay; it is not obtained by merely extending the current regression field list.

Thus the induction theorem is not an arbitrary-redundancy classification, the redundancy-eight result is a high-field theorem, and bounded-field and modular-carrier strata remain open. In particular, the paper does not settle the general Reed–Solomon deep-hole conjecture.

Two companion papers will take up the branches deliberately left out here. The first begins with the four-marker redundancy-nine problem and its binary-quartic slices; it will state no classification until the characteristic-five carrier and the component exhaustion are reconciled. The second treats the logically separate ordered-Hessian, Lucas-carrier, and distinguished e_7 geometry. These forthcoming analyses are not inputs to any theorem in the present paper.

The structural conclusion is nevertheless stable. Beyond redundancy four, deepness is no longer controlled by one invariant or by isolated bad fibers. The persistent geometry is a coherent rank-two polar flag; the modular geometry is a Lucas-computed contraction kernel; and the remaining arithmetic is the Frobenius action on a normalized ordered-root cover. Redundancy five is the first exceptional-cover case, while the later theorems show how that cover is propagated, contained, or replaced in the levels and carrier strata treated here.

Appendix

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